

SIMATS ENGINEERING ASSESSMENT TOOL 2

**COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713**

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

*I Harish Kumar / 192521340 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Scenario-Based

1. Answer:

a) Network & broadcast addresses: School of Computing (/25) – Network: 192.168.100.0, Broadcast: 192.168.100.127. School of Electronics (/26) – Network: 192.168.100.128, Broadcast: 192.168.100.191. School of Mechanical Engineering (/27) – Network: 192.168.100.192, Broadcast: 192.168.100.223.

b) Valid host IP range: Computing: 192.168.100.1–192.168.100.126. Electronics: 192.168.100.129–192.168.100.190. Mechanical: 192.168.100.193–192.168.100.222.

c) Usable host addresses: Computing: 126 hosts. Electronics: 62 hosts. Mechanical: 30 hosts.

d) Remaining unused range: 192.168.100.224–192.168.100.255 (32 addresses, an unused /27-sized block).

2. Answer:

a) Network & broadcast addresses: Software Development (/24) – Network: 10.10.0.0, Broadcast: 10.10.0.255. Quality Assurance (/25) – Network: 10.10.1.0, Broadcast: 10.10.1.127. Human Resources (/26) – Network: 10.10.1.128, Broadcast: 10.10.1.191. Executive Management (/27) – Network: 10.10.1.192, Broadcast: 10.10.1.223.

b) Valid host IP range: Software Development: 10.10.0.1–10.10.0.254. Quality Assurance: 10.10.1.1–10.10.1.126. Human Resources: 10.10.1.129–10.10.1.190. Executive Management: 10.10.1.193–10.10.1.222.

c) Usable hosts: Software Development: 254. Quality Assurance: 126. Human Resources: 62. Executive Management: 30.

d) Unused IP address range remaining: 10.10.1.224–10.10.255.255 (65,056 addresses left unallocated for future departments).

3. Answer:

a) Compressed notation: 2001:db8:abcd::/64

b) Expanded (full) representation: 2001:0db8:abcd:0000:0000:0000:0000:0000

c) Network prefix (/64): 2001:0db8:abcd:0000::/64 (i.e. 2001:db8:abcd::/64 compressed)

d) Total host addresses supported: 2^{64} ($128 - 64 = 64$ host bits) $\approx 1.8 \times 10^{19}$ addresses.

e) Total possible host addresses in this network: Same as above, 2^{64} , since this asks the same thing as part (d).

4. Answer:

a) Destination subnet: 192.168.2.0/24 (since 192.168.2.45 falls in the range 192.168.2.0–192.168.2.255).

b) Matching routing table entry: 192.168.2.0/24

c) Next-hop network/interface: The interface directly connected to (configured for) the 192.168.2.0/24 subnet.

d) Default route fallback: If no entry matched, the router would use its default route 0.0.0.0/0, forwarding the packet to its configured default gateway/upstream ISP link.

5. Answer:

a) Network & broadcast addresses: Administration LAN (192.168.10.0/26) – Network: 192.168.10.0, Broadcast: 192.168.10.63. Finance LAN (192.168.10.64/26) – Network: 192.168.10.64, Broadcast: 192.168.10.127.

b) Valid host IP range: Administration: 192.168.10.1–192.168.10.62. Finance: 192.168.10.65–192.168.10.126.

c) Suggested computer IP addresses: Administration: 192.168.10.2, 192.168.10.3, 192.168.10.4. Finance: 192.168.10.66, 192.168.10.67, 192.168.10.68.

d) Default gateway: Administration LAN hosts use 192.168.10.1. Finance LAN hosts use 192.168.10.65.